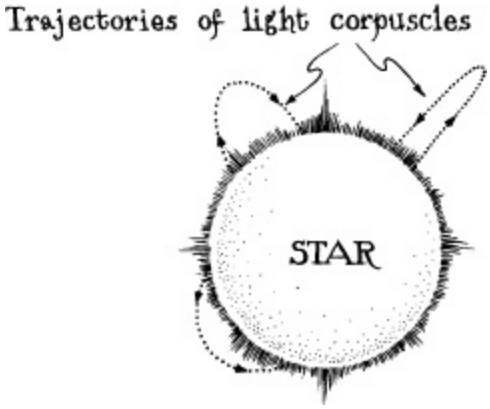
Michell could easily compute the critical circumference; it was 18.5 kilometers, if the star had the same mass as the Sun, and proportionately larger if the mass were larger.



3.1 The behavior of light emitted from a star that is smaller than the critical circumference, as computed in 1783 by John Michell using Newton's laws of gravity and corpuscular description of light.

Nothing in the eighteenth-century laws of physics prevented so compact a star from existing. Thus, Michell was led to speculate that the Universe might contain a huge number of such dark stars, each living happily inside its own critical circumference, and each invisible from Earth because the corpuscles of light emitted from its surface are inexorably pulled back down. Such *dark stars* were the eighteenth-century versions of black holes.

Michell, who was Rector of Thornhill in Yorkshire, England, reported his prediction that dark stars might exist to the Royal Society of London on 27 November 1783. His report made a bit of a splash among British natural philosophers. Thirteen years later, the French natural philosopher Pierre Simon Laplace popularized the same prediction in the first edition of his famous work *Le Systeme du Monde,* without reference to Michell's earlier work. Laplace kept his dark-star prediction in the second (1799) edition, but by the time of the third (1808) edition, Thomas Young's discovery of the interference of light with itself[1] was forcing natural philosophers to abandon the corpuscular description of light in favor of a wave description devised by Christiaan Huygens—and it was not at all clear how this wave description should be meshed with Newton's laws of gravity so as to compute the effect of a star's gravity on the light it emits. For this reason, presumably, Laplace deleted the concept of a dark star from the third and subsequent editions of his book.